

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION  
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET G — FINAL MASTERY & GRAND CHALLENGE TEST (60 MINS)

CHAPTER 05: EXPLORING MIXTURES: SOLUTION, COLLOID & SUSPENSION  
TIME ALLOWED: 60 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-G-05

Student's Name: \_\_\_\_\_

Roll No.: \_\_\_\_\_

Class & Section: Class IX - \_\_\_\_\_

Date of Exam: \_\_\_\_\_

Invigilator's Sign: \_\_\_\_\_

GENERAL INSTRUCTIONS (SET G - FINAL MASTERY & GRAND CHALLENGE TEST):

1. The question paper comprises **25 High-Tier Mastery & Olympiad-Standard Multiple Choice Questions** of **1 mark each**.
2. Tests **In-Depth Conceptual Mastery, Multi-Step Numericals & Rigorous Scientific Deductions**: Recommended target time is **60 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **GRAND CHALLENGE TIER:** This paper features comprehensive multi-step numericals, advanced Olympiad/NTSE-aligned scenarios, and rigorous experimental synthesis designed to achieve absolute 100/100 mastery.

**Q.1** What is the mass of pure water required to dissolve 50 g of sodium chloride ( $\text{NaCl}$ ) to produce a 20% (by mass) solution?

- (A) 200 g (B) 250 g  
(C) 150 g (D) 100 g

**Q.2** Which of the following is classified as a sol rather than a gel?

- (A) Cheese (B) Jelly  
(C) Paints (D) Butter

**Q.3** During fractional distillation of liquid air, which gas boils off first due to having the lowest boiling point?

- (A) Nitrogen (boiling point  $-196^\circ\text{C}$ ) (B) Argon (boiling point  $-186^\circ\text{C}$ )  
(C) Oxygen (boiling point  $-183^\circ\text{C}$ ) (D) Carbon dioxide (boiling point  $-78^\circ\text{C}$ )

**Q.4** A mixture containing 5.0 g of sand, 3.0 g of common salt ( $\text{NaCl}$ ), and 2.0 g of ammonium chloride ( $\text{NH}_4\text{Cl}$ ) can be separated into pure components by:

- (A) Sublimation  $\rightarrow$  Dissolution in water & filtration  $\rightarrow$  Evaporation/Crystallization  
(B) Filtration  $\rightarrow$  Distillation  $\rightarrow$  Chromatography  
(C) Magnetic separation  $\rightarrow$  Sublimation  $\rightarrow$  Centrifugation  
(D) Fractional distillation directly

**Q.5** Why does a colloidal sol of ferric hydroxide ( $\text{Fe}(\text{OH})_3$ ) coagulate (precipitate) upon addition of sodium sulfate ( $\text{Na}_2\text{SO}_4$ )?

- (A) Sulfate anions ( $\text{SO}_4^{2-}$ ) neutralize the positive zeta-potential charge on the colloidal particles (Hardy-Schulze rule) (B) Sulfate absorbs all water  
(C) Iron dissolves completely (D) Temperature drops to absolute zero

**Q.6 How does the solubility of calcium oxide ( $\text{CaO}$ ) in water behave with increasing temperature?**

- (A) Increases rapidly
- (B) Decreases, because dissolution of calcium oxide/hydroxide is highly exothermic
- (C) Remains strictly constant
- (D) Becomes infinite

**Q.7 What is the Retention Factor ( $R_f$ ) of a solute that does not dissolve in the mobile solvent and remains stationary at the baseline?**

- (A) 1.0
- (B) Zero (0.0)
- (C) 0.5
- (D) Infinity

**Q.8 Which of the following processes represents an endothermic physical change?**

- (A) Freezing of water
- (B) Sublimation of dry ice (solid  $\text{CO}_2$ )
- (C) Condensation of steam
- (D) Rusting of iron

**Q.9 A saturated solution of sugar at  $80^\circ\text{C}$  contains 360 g of sugar per 100 g of water. At  $20^\circ\text{C}$ , solubility is 200 g per 100 g of water. If 230 g of the saturated solution at  $80^\circ\text{C}$  is cooled to  $20^\circ\text{C}$ , what mass of sugar crystals precipitates?**

- (A) 160 g
- (B) 80 g
- (C) 100 g
- (D) 50 g

**Q.10 Which optical phenomenon proves that milk is a colloidal dispersion rather than a true solution?**

- (A) Total internal reflection
- (B) Tyndall scattering
- (C) Refraction of light
- (D) Dispersion into a rainbow spectrum

**Q.11 What is the primary role of the fractionating column packed with glass beads in fractional distillation?**

- (A) To filter impurities
- (B) To provide a large surface area for repeated cycles of condensation and revaporization, sharpening component separation
- (C) To react chemically with vapors
- (D) To cool the boiling flask

**Q.12 Which of the following is classified as an emulsion?**

- (A) Cod liver oil in water
- (B) Shaving cream
- (C) Pumice stone
- (D) Smoke

**Q.13 A mixture of acetone (boiling point  $56^\circ\text{C}$ ) and water (boiling point  $100^\circ\text{C}$ ) is best separated by:**

- (A) Simple distillation
- (B) Separating funnel
- (C) Crystallization
- (D) Filtration

**Q.14 Which property indicates that air is a mixture rather than a chemical compound?**

- (A) Its composition varies with altitude and location
- (B) Its components retain their individual chemical properties
- (C) It can be separated by physical methods (liquefaction and fractional distillation)
- (D) All of the above

**Q.15 What is the mass percentage of solute in a solution prepared by dissolving 25 g of glucose in 175 g of water?**

- (A) 12.5%
- (B) 14.3%
- (C) 25.0%
- (D) 10.0%

**Q.16** Which state of matter has neither definite shape nor definite volume, and consists of ionized gas at extremely high temperatures?

- (A) Solid
- (B) Liquid
- (C) Gas
- (D) Plasma

**Q.17** Bose-Einstein Condensate (BEC) is formed by:

- (A) Heating gas to millions of degrees
- (B) Cooling a gas of extremely low density to ultra-low temperatures near absolute zero
- (C) Compressing liquid water to 10,000 atmospheres
- (D) Subjecting solid ice to high voltage

**Q.18** Which gas is evolved when pure iron sulfide (FeS) reacts with dilute sulfuric acid ( $\text{H}_2\text{SO}_4$ )?

- (A)  $\text{H}_2$
- (B)  $\text{H}_2\text{S}$  (Hydrogen sulfide)
- (C)  $\text{SO}_2$
- (D)  $\text{O}_2$

**Q.19** Why does a diamond have an extremely high melting point ( $\sim 3500^\circ\text{C}$ ) even though carbon is a non-metal?

- (A) It has metallic bonding
- (B) It consists of a 3D giant covalent network lattice where each carbon is tetrahedrally bonded to four others by strong covalent bonds
- (C) It contains ionic calcium
- (D) It is magnetic

**Q.20** Which separation method separates blood proteins by utilizing an electric field?

- (A) Centrifugation
- (B) Gel Electrophoresis
- (C) Chromatography
- (D) Sublimation

**Q.21** What is the physical state of the dispersed phase and dispersion medium in butter?

- (A) Dispersed phase: Liquid (water); Dispersion medium: Solid (fat)
- (B) Dispersed phase: Solid; Dispersion medium: Liquid
- (C) Both are gases
- (D) Both are solids

**Q.22** Which of the following represents a purely physical change?

- (A) Sublimation of camphor
- (B) Digestion of starch by amylase
- (C) Electrolysis of water into  $\text{H}_2$  and  $\text{O}_2$
- (D) Burning of magnesium ribbon

**Q.23** If 10 mL of alcohol is dissolved in 90 mL of water, the volume percentage concentration of alcohol is:

- (A) 10%
- (B) 11.1%
- (C) 9%
- (D) 100%

**Q.24** Why is fractional crystallization effective for separating two soluble salts with different solubility curves?

- (A) Because the less soluble salt crystallizes out first upon cooling, leaving the more soluble salt in the mother liquor
- (B) Because one salt evaporates
- (C) Because one salt becomes a gas
- (D) Because of magnetic fields

**Q.25** Which colloid has solid as the dispersed phase and solid as the dispersion medium?

- (A) Ruby glass / colored gemstones (Solid sol)
- (B) Cheese
- (C) Pumice
- (D) Smoke

**CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET G)**

|    |                                                                                                 |    |                                                                                                 |    |                                                                                                 |    |                                                                                                 |    |                                                                                                 |
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Total Questions Attempted: \_\_\_\_ / 25

Marks Obtained: \_\_\_\_ / 25

Evaluator's Signature: \_\_\_\_\_

### OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET G)

| Q. No. | Correct | Rigorous Scientific Solution & Formula Derivation                                                                                                                                                                                                                                                |
|--------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q.1    | (A)     | $\text{Mass \%} = \frac{m_{\text{solute}}}{m_{\text{solute}} + m_{\text{water}}} \times 100 \rightarrow 20 = \frac{50}{50 + m_{\text{water}}} \times 100 \rightarrow 50 + m_{\text{water}} = 250 \rightarrow m_{\text{water}} = 200 \text{ g}$ .                                                 |
| Q.2    | (C)     | Paints are sols (solid pigment particles dispersed in liquid vehicle). Cheese, jelly, and butter are gels (liquid trapped in solid).                                                                                                                                                             |
| Q.3    | (A)     | Nitrogen has the lowest boiling point ( $-196^\circ\text{C}$ ) and distills off first at the top of the fractionating column.                                                                                                                                                                    |
| Q.4    | (A)     | 1. Sublimation separates $\text{NH}_4\text{Cl}$ . 2. Dissolving remainder in water and filtering leaves sand on filter paper. 3. Evaporating water leaves pure salt.                                                                                                                             |
| Q.5    | (A)     | According to the Hardy-Schulze rule, oppositely charged polyvalent ions ( $\text{SO}_4^{2-}$ ) neutralize colloidal surface charges, inducing coagulation.                                                                                                                                       |
| Q.6    | (B)     | Dissolution of $\text{CaO}$ and $\text{Ca(OH)}_2$ is strongly exothermic; by Le Chatelier's principle, adding heat decreases solubility.                                                                                                                                                         |
| Q.7    | (B)     | $R_f = \frac{\text{Distance travelled by solute}}{\text{Distance travelled by solvent front}} = \frac{0}{d_{\text{solvent}}} = 0.0$ .                                                                                                                                                            |
| Q.8    | (B)     | Sublimation requires absorption of latent heat to break intermolecular bonds without changing chemical identity.                                                                                                                                                                                 |
| Q.9    | (B)     | At $80^\circ\text{C}$ , total mass of solution $= 360 + 100 = 460 \text{ g}$ . A 230 g sample contains half the amounts: $180 \text{ g}$ sugar and $50 \text{ g}$ water. At $20^\circ\text{C}$ , 50 g of water holds $100 \text{ g}$ sugar. Precipitated crystals $= 180 - 100 = 80 \text{ g}$ . |
| Q.10   | (B)     | The Tyndall effect illuminates the light path through milk, proving the presence of colloidal fat and protein micelles.                                                                                                                                                                          |
| Q.11   | (B)     | The column packing creates a thermal gradient where rising vapor undergoes multiple theoretical condensation-vaporization plates.                                                                                                                                                                |
| Q.12   | (A)     | Cod liver oil dispersed in water is an emulsion (liquid dispersed in liquid).                                                                                                                                                                                                                    |
| Q.13   | (A)     | Acetone and water are miscible with a boiling point difference $> 25^\circ\text{C}$ ( $44^\circ\text{C}$ difference), suitable for simple distillation.                                                                                                                                          |
| Q.14   | (D)     | Variable composition, retention of individual constituent properties, and physical separability are hallmarks of a mixture.                                                                                                                                                                      |
| Q.15   | (A)     | $\text{Mass \%} = \frac{25}{25 + 175} \times 100 = \frac{25}{200} \times 100 = 12.5\%$ .                                                                                                                                                                                                         |
| Q.16   | (D)     | Plasma is the fourth state of matter consisting of superheated ionized gas and free electrons (found in stars, lightning, neon tubes).                                                                                                                                                           |
| Q.17   | (B)     | BEC forms when a low-density boson gas is cooled to near absolute zero ( $< 10^{-6} \text{ K}$ ), collapsing atoms into a single quantum state.                                                                                                                                                  |
| Q.18   | (B)     | $\text{FeS} + \text{H}_2\text{SO}_4 \rightarrow \text{FeSO}_4 + \text{H}_2\text{S} \uparrow$ . Hydrogen sulfide smells like rotten eggs.                                                                                                                                                         |
| Q.19   | (B)     | Diamond's rigid 3D covalent network requires massive thermal energy to break strong $\text{C-C}$ covalent bonds.                                                                                                                                                                                 |
| Q.20   | (B)     | Gel electrophoresis separates charged biological macromolecules (proteins, DNA) based on molecular size and charge.                                                                                                                                                                              |
| Q.21   | (A)     | Butter is a water-in-oil emulsion / gel where liquid water droplets are dispersed within solid fat matrix.                                                                                                                                                                                       |
| Q.22   | (A)     | Sublimation changes only the physical state from solid to vapor without altering chemical composition.                                                                                                                                                                                           |
| Q.23   | (A)     | $\text{Volume \%} = \frac{10}{10 + 90} \times 100 = \frac{10}{100} \times 100 = 10\%$ .                                                                                                                                                                                                          |

| Q. No. | Correct | Rigorous Scientific Solution & Formula Derivation                                                                                        |
|--------|---------|------------------------------------------------------------------------------------------------------------------------------------------|
| Q.24   | (A)     | Fractional crystallization exploits differential temperature-solubility coefficients to selectively harvest separate crystalline phases. |
| Q.25   | (A)     | Colored gemstone glasses and minerals are solid sols (solid particles dispersed in solid matrix).                                        |